📊 OpenTelemetry Metrics Analysis Report

Comprehensive Infrastructure & Kubernetes Metrics Assessment

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Executive Summary

This report provides a comprehensive analysis of OpenTelemetry metrics instrumentation across your Kubernetes infrastructure. The analysis covers 23,621 metric data points from 36 unique metrics across 7 namespaces, collected over a 14-minute observation period.

Overall Metrics Score

80/100

Status: GOOD

📈 Key Metrics Overview

Total Metric Data Points
23,621

Unique Metrics
36

Namespaces Monitored
7

Data Collection Period
14 min

🎯 Metrics Coverage Assessment

Infrastructure Metrics

Category	Metrics Detected	Coverage	Status
CPU Metrics	system.cpu.time, system.cpu.load_average.{1m,5m,15m}	Excellent	✓ Complete
Memory Metrics	system.memory.usage	Good	✓ Present
Network Metrics	system.network.{io,errors,dropped,packets,connections}	Excellent	✓ Complete
Filesystem Metrics	system.filesystem.{usage,inodes.usage}	Good	✓ Present

Kubernetes Metrics

Category	Metrics Detected	Coverage	Status
Pod CPU	k8s.pod.cpu.{time,utilization}	Excellent	✓ Complete
Pod Memory	k8s.pod.memory.{usage,working_set,rss,available,page_faults,major_page_faults}	Excellent	✓ Complete
Pod Network	k8s.pod.network.{io,errors}	Good	✓ Present
Pod Filesystem	k8s.pod.filesystem.{capacity,usage,available}	Good	✓ Present
Container Metrics	container.{cpu,memory,filesystem}.*	Excellent	✓ Complete

📊 Top Metrics by Volume

Metric Name	Count	Percentage	Category
system.cpu.time	2,880	12.2%	System
k8s.pod.network.errors	1,288	5.5%	Kubernetes
k8s.pod.network.io	1,288	5.5%	Kubernetes
system.network.{dropped,packets,io,errors}	3,528	14.9%	System
Container metrics (all)	5,836	24.7%	Container
Pod metrics (all)	7,084	30.0%	Kubernetes

🏢 Namespace Distribution

Namespace	Metric Count	Unique Metrics	Percentage
kube-system	8,712	24	36.9%
(null/system)	8,244	12	34.9%
otel-gateway	2,288	24	9.7%
core-services	1,498	24	6.3%
otelagent	1,129	24	4.8%
observability	1,056	22	4.5%
local-path-storage	1,056	22	4.5%
ingress-nginx	1,056	22	4.5%

🏷️ Attribute Quality Assessment

Most Used Attributes

Attribute Key	Usage Count	Purpose	Quality
direction	6,368	Network I/O direction (transmit/receive)	Excellent
device	4,617	Block device identification	Excellent
state	4,479	CPU/Memory state classification	Excellent
cpu	2,976	CPU core identification	Excellent
interface	2,696	Network interface identification	Excellent

Cardinality Analysis

system.cpu.time cardinality: 96 unique combinations

- 8 CPU states: idle, user, system, wait, interrupt, softirq, steal, nice
- Multiple CPU cores per node
- ✓ **Well-managed cardinality**

✅ Strengths

Excellent OTel Semantic Convention Compliance

Your metrics follow OpenTelemetry semantic conventions perfectly with proper naming patterns (system.*, k8s.*, container.*). This ensures compatibility with OTel-aware tools and dashboards.

Comprehensive Infrastructure Coverage

Strong coverage of system-level metrics including CPU, memory, network, and filesystem. All critical infrastructure components are being monitored.

Well-Managed Cardinality

Attribute usage shows controlled cardinality with 96 combinations for CPU metrics across 8 states. This prevents metric explosion while maintaining useful dimensionality.

Rich Kubernetes Context

Excellent Kubernetes observability with comprehensive pod, container, and namespace metrics. Container and pod lifecycle metrics are well-instrumented.

⚠️ Areas for Improvement

CRITICAL: Missing Application-Level Metrics

Only infrastructure and Kubernetes metrics detected. No application performance metrics found such as:

- HTTP request rates and latencies (http.server.request.duration)
- Database query metrics (db.client.operation.duration)
- Business KPIs and custom metrics
- Service-level metrics (error rates, throughput)

Recommendation: Instrument your applications with OTel SDKs to capture application-specific metrics.

CRITICAL: Empty Resource Attributes

Metrics have empty resource objects, missing critical service identification attributes:

- service.name
- service.version
- deployment.environment
- service.instance.id

Recommendation: Configure resource detection processor in OTel Collector to populate these attributes.

WARNING: 8,244 Metrics with NULL Namespace

34.9% of metrics have NULL namespace values, making it difficult to attribute metrics to specific services or components.

Recommendation: Ensure all metrics are properly tagged with namespace context, particularly system-level metrics.

WARNING: No Custom/Business Metrics

All metrics are infrastructure-focused. Consider adding business-relevant metrics such as:

- Order processing rates
- User activity metrics
- Transaction success rates
- Feature usage statistics

🎯 Priority Recommendations

Priority	Recommendation	Impact	Effort
P0	Add application-level instrumentation (HTTP, DB, RPC)	High	Medium
P0	Populate resource attributes (service.name, etc.)	High	Low
P1	Fix NULL namespace values for system metrics	Medium	Low
P1	Add custom business metrics	Medium	Medium
P2	Implement metric aggregation for long-term storage	Low	Medium

📋 Detailed Scoring Breakdown

Category	Score	Weight	Weighted Score
Metrics Coverage	75/100	25%	18.75
Metrics Quality	85/100	25%	21.25
Naming Conventions	90/100	15%	13.50
Cardinality Management	80/100	15%	12.00
Data Completeness	60/100	20%	12.00
Overall Score	**Weighted Average**		**77.5/100**

For questions or additional analysis, please contact your observability team